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Problem Set 2.4.

1. $B_{5\times3} \cdot A_{3\times5}$ is allowed and result will be $C_{5\times5}$.

$A_{3\times5} \cdot B_{5\times3}$ is allowed and result will be $C_{3\times3}$.

$ABD = A_{3\times5} \cdot B_{5\times3} \cdot D_{3\times1}$ ~~(Y) → C_1~~ NO

$D_{3\times1} \cdot C_{5\times1}$ is NO

$A(B+C) = A_{3\times5}(\underbrace{B_{5\times3} + C_{5\times1}}_{\text{cannot be added.}})$ NO

2. a) all rows of A with 2nd column of B (3rd method)

OR $A \cdot \vec{b_2} \rightarrow 2^{nd}$ method of matrix multiplication

b) row 1 of A multiplied by B (3^{rd} method)

c) $c_{35} = \sum_{i=1}^{m} a_{3i} \cdot b_{i5}$ (1^{st} method)

d) CDE : - multiply [row 1 of C] · D ⇒ row 1 of CD $\overset{not}{=} x_{1i}$

- multiply : $y_{11} = \sum_{j=1}^{m} x_{1i} \cdot e_{i1}$.

3. $AB = \begin{bmatrix} 1 & 5 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 0 & 2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 7 \\ 0 & 7 \end{bmatrix}$

$AC = \begin{bmatrix} 1 & 5 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 6 & 2 \end{bmatrix}$

$A(B+C) = \begin{bmatrix} 1 & 5 \\ 2 & 3 \end{bmatrix} \left(\begin{bmatrix} 0 & 2 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 3 & 1 \\ 0 & 0 \end{bmatrix} \right) = \begin{bmatrix} 1 & 5 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 3 & 3 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 8 \\ 6 & 9 \end{bmatrix}$

OBS: $AB + AC = A(B+C)$

4. $A(BC) = \begin{bmatrix} 1 & 5 \\ 2 & 3 \end{bmatrix} \left(\begin{bmatrix} 0 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 0 & 0 \end{bmatrix} \right) = \begin{bmatrix} 1 & 5 \\ 2 & 3 \end{bmatrix} \cdot \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = 0_{2\times2}$.

$(AB) \cdot C = \left(\begin{bmatrix} 1 & 5 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 0 & 2 \\ 0 & 1 \end{bmatrix} \right) \begin{bmatrix} 3 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 7 \\ 0 & 7 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 0 & 0 \end{bmatrix} = 0_{2\times2}$.

5.

$$A^2 = \begin{bmatrix} 1 & b \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & b \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2b \\ 0 & 1 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 1 & b \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 2b \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 3b \\ 0 & 1 \end{bmatrix}$$

$\vdots$

$$A^n = \begin{bmatrix} 1 & n \cdot b \\ 0 & 1 \end{bmatrix}$$; for A^5 replace n with 5.

$$A^2 = \begin{bmatrix} 2 & 2 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} 2 & 2 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 4 & 4 \\ 0 & 0 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 2 & 2 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} 4 & 4 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 8 & 8 \\ 0 & 0 \end{bmatrix}$$

$\vdots$

$$A^n = \begin{bmatrix} 2^n & 2^n \\ 0 & 0 \end{bmatrix}$$

6.

$$A + B = \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 3 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 2 \\ 3 & 0 \end{bmatrix}$$

$$(A+B)^2 = \begin{bmatrix} 2 & 2 \\ 3 & 0 \end{bmatrix}\begin{bmatrix} 2 & 2 \\ 3 & 0 \end{bmatrix} = \begin{bmatrix} 10 & 4 \\ 6 & 6 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}$$

$$B^2 = \begin{bmatrix} 1 & 0 \\ 3 & 0 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 3 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 3 & 0 \end{bmatrix}$$

$$AB = \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 3 & 0 \end{bmatrix} = \begin{bmatrix} 7 & 0 \\ 0 & 0 \end{bmatrix}$$

$$A^2 + 2AB + B^2 = \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 14 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 3 & 0 \end{bmatrix} = \begin{bmatrix} 16 & 2 \\ 3 & 0 \end{bmatrix} \neq (A+B)^2$$

7. ~~a) FALSE~~ $\cancel{\begin{bmatrix}0&1\\1&0\end{bmatrix}\begin{bmatrix}1&2\\3&4\end{bmatrix}=\begin{bmatrix}3&4\\1&2\end{bmatrix}}$ → ~~not the same columns.~~

~~$[A\ 2\ 3]$~~

~~b) False~~

a) True → Columns of AB are computed as:

$$\left[A\vec{b}_1 \quad A\vec{b}_2 \quad A\vec{b}_3 \quad \dots \quad A\vec{b}_p\right]$$

If $\vec{b}_1 = \vec{b}_3 \Rightarrow A\vec{b}_1 = A\vec{b}_3$.

b) False $\begin{bmatrix}1&-1&0\\0&1&0\\0&0&1\end{bmatrix}\begin{bmatrix}2&2&2\\1&2&3\\2&2&2\end{bmatrix}=\begin{bmatrix}1&0&-1\\1&2&3\\2&2&2\end{bmatrix}$

c) AB can be computed as $\begin{bmatrix}(\text{row } 1 \text{ of } A)\cdot B\\(\text{row } 2 \text{ of } A)\cdot B\\(\text{row } 3 \text{ of } A)\cdot B\\\vdots\\(\text{row } m \text{ of } A)\cdot B\end{bmatrix}$

If row 1 = row 3 then $\underbrace{(\text{row } 1 \text{ of } A)\cdot B = (\text{row } 3 \text{ of } A)\cdot B}$

$\Downarrow$

ABC will have rows 1 and 3 equal.

d) $(AB)^2 = A^2B^2$

$(AB)^2 = AB\cdot AB$

$A^2B^2 = AA\cdot BB$ ⇒ False

$A=\begin{bmatrix}1&-1&0\\0&1&0\\0&0&1\end{bmatrix}$

$B=\begin{bmatrix}1&0&0\\0&1&-1\\0&0&1\end{bmatrix}$

8.

$$DA = \begin{bmatrix} 3 & 0 \\ 0 & 5 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 3a & 3b \\ 5c & 5d \end{bmatrix}$$

row 1 is multiplied by 3

row 2 is multiplied by 5.

$$EA = \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} c & d \\ c & d \end{bmatrix}$$

row 1 is replaced with row 2.

$$AD = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & 5 \end{bmatrix} = \begin{bmatrix} 3a & 5b \\ 3c & 5d \end{bmatrix}$$

~~row~~ col 1 is multiplied by 3

col 2 is multiplied by 5.

$$AE = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & a+b \\ 0 & c+d \end{bmatrix}$$ col 1 is 0

col 2 is col 1 + col 2.

9. a)

$$AF = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} a & a+b \\ c & c+d \end{bmatrix}$$

$$E(AF) = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} a & a+b \\ c & c+d \end{bmatrix} = \begin{bmatrix} a & a+b \\ a+c & a+b+c+d \end{bmatrix}$$

b) $(EA)F = E(AF)$ ⇒ associativity of matrix multiplication is obeyed.

10.

$$FA = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a+c & b+d \\ c & d \end{bmatrix}$$

$$E(FA) = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} FA = \begin{bmatrix} a+c & b+d \\ a+2c & b+2d \end{bmatrix}$$

$E(FA) \neq FEA$ ⇒ commutativity is NOT obeyed.

11.

~~R(AC) = CAR~~

$(RA)C = R(AC)$; R = row operation

C = column operation.

This is true thanks to the associativity of matrix multiplication.

12.

a) $B = 4 \cdot I$; $\begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix}$

b) $B = 0$; $B = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

c) $B = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$

d) $B = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}$

13.

$$AB = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} a & 0 \\ c & 0 \end{bmatrix}$$

$$BA = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a & b \\ 0 & 0 \end{bmatrix}$$

; For $AB = BA \Rightarrow c = b = 0 \; \forall a \in \mathbb{R}$

$$AC = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & a \\ 0 & c \end{bmatrix}$$

$$CA = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} c & d \\ 0 & 0 \end{bmatrix}$$

; For $AC = CA \Rightarrow c = 0$ and $a = d \in \mathbb{R}$

$$\text{For} \begin{cases} AB = BA \\ \& \\ AC = CA \end{cases} \Rightarrow \begin{cases} a = d \\ b = c = 0. \end{cases}$$

14.

$$(A-B)^2 = (A^{\cancel{2}} - B)\cdot(A-B) = \cancel{(A-B)A}\; A(A-B)\cancel{A} - B(A-B) =$$

$$= \boxed{A^2 - AB - BA + B^2.}$$

$\boxed{(B-A)^2} = (A-B)^2$

$A^2 - AB - BA + B^2$

$$(B-A)(B-A) = B(B-A) - A(B-A) = B^2 - BA - AB + A^2 \; ✓$$

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15. a) If $A_{m \times n}$ then $A_{m \times n} \cdot A_{m \times n}$ cannot be multiplied

$\Rightarrow$ statement is True

b) False $A_{m \times n} \cdot B_{n \times m}$, $B_{n \times m} \cdot A_{m \times n}$ are not squared and can be multiplied.

c) True see b).

d) True

16. a) $A_{m \times n} \cdot \vec{x}_n \Rightarrow m \cdot n$ multiplications

b) $A_{m \times n} \cdot B_{n \times p} \Rightarrow m \cdot n \cdot p$ multiplications

Each $c_{ij} = \underbrace{\sum_{x=1}^{n} a_{ix} b_{xj}}$

n multiplications for 1 element but $C = AB$ is of size $m \times p$.

c) n^3 multiplications

2nd method

17. a) col 2 of AB: $\begin{bmatrix} 2 \\ 3 \end{bmatrix} \cdot 0 + \begin{bmatrix} -1 \\ -2 \end{bmatrix} \cdot 0 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

3rd method

b) row 2 of AB: $\begin{bmatrix} 3 & -2 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 4 \\ 1 & 0 & 6 \end{bmatrix} = \begin{bmatrix} 3 & 0 & 0 \end{bmatrix}$

c) row 2 of $A^2 = \begin{bmatrix} 3 & -2 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ 3 & -2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \end{bmatrix}$

d) row 2 of $A^3 = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ 3 & -2 \end{bmatrix} = \begin{bmatrix} 3 & -2 \end{bmatrix}$

18. a) $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{bmatrix}$ b) $A = \begin{bmatrix} 1 & -1 & 1 \\ -1 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix}$ c) $A = \begin{bmatrix} 1 & 0{,}5 & 1/3 \\ 2 & 1 & 2/3 \\ 3 & 1{,}5 & 1 \end{bmatrix}$

19. a) $a_{ij} = 0 \ \forall\, i \neq j \Rightarrow$ matrices that keep the same rows and just multiply them.

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$\Downarrow$ diagonal matrix.

b) $$\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \rightarrow \text{Lower triangular matrix}$$

c) $$\begin{bmatrix} 1 & 2 & 4 \\ 2 & 3 & 5 \\ 4 & 5 & 6 \end{bmatrix} \rightarrow \text{Symetric matrix}$$

d) $$\begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ 1 & 2 & 3 \end{bmatrix} \rightarrow \text{All rows equal matrix}$$

20. a) a_{11} ; b) $\ell_{31} = +\dfrac{a_{31}}{a_{11}}$; c) $a'_{32} = a_{32} - \dfrac{a_{31}}{a_{11}} \cdot a_{12}$

d) $a'_{22} = a_{22} - \dfrac{a_{21}}{a_{11}} a_{12}$

21.
$$A^2 = \begin{bmatrix} 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 0 & 0 & 0 & 8 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} ; A^4 = O_{4\times4}.$$

$$A\vec{v} = \begin{bmatrix} 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ t \end{bmatrix} = \begin{bmatrix} 2y \\ 2z \\ 2t \\ 0 \end{bmatrix} ; A^2\vec{u} = \begin{bmatrix} 4z \\ 4t \\ 0 \\ 0 \end{bmatrix} ; A^3\vec{u} = \begin{bmatrix} 8t \\ 0 \\ 0 \\ 0 \end{bmatrix} ; A^4\vec{u} = O_{1\times4}$$

22.

$A^2 = -I$; $-I = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$; $A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$

$\cancel{\begin{bmatrix} 1 & -2 \\ -2 & 1 \end{bmatrix}\begin{bmatrix} 1 & \\ -2 & 1 \end{bmatrix}} \quad \cancel{\begin{bmatrix} -2 & 1 \\ -1 & \end{bmatrix}\begin{bmatrix} & -1 \\ -1 & \end{bmatrix}}$

$BC = 0$; $C = \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix}$; $B = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

$DE = -ED$ => Comutativity is allowed only for multiples of $I \Rightarrow E = xI$; $x \in \mathbb{R}$.

$\begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} -2 & 0 \\ 0 & -1 \end{bmatrix}$

OR not ...

$\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}\begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} -2 & 0 \\ 0 & -1 \end{bmatrix}$

See solutions for <u>trial and error</u> matrices :)

23. a) $A^2 = 0$

$A = \begin{bmatrix} 0 & 0 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ or $\cancel{A = \begin{bmatrix} 0 & 1 & -1 \\ 1 & 0 & 0 \end{bmatrix}}$ $\cancel{A = \begin{bmatrix} 1 & 1 & -1 \\ 0 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix}}$

$\cancel{A = \begin{bmatrix} 1 & -1 & 1 \\ 0 & & \end{bmatrix}}$ $\cancel{A = \begin{bmatrix} 0 & 1 & -1 \\ 0 & 1 & 1 \end{bmatrix}}$ $A = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 1 & -1 \end{bmatrix}$.

b) $A = \begin{bmatrix} 0 & 2 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix}$

24.

$A_1 = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$; $A_1^2 = \begin{bmatrix} 4 & \overset{2+1}{3} \\ 0 & 1 \end{bmatrix}$; $A_1^3 = \begin{bmatrix} 8 & \overset{4\cdot 3+3\cdot 1}{12+3} \\ 0 & 1 \end{bmatrix}$

$A_1^n = \begin{bmatrix} 2^n & 2^{n-1}\cdot\left(a_{12}^{(n-1)}+1\right) \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2^n & 2^n - 1 \\ 0 & 1 \end{bmatrix}$

$$A_2^2 = \begin{bmatrix}1 & 1\\ 1 & 1\end{bmatrix}\begin{bmatrix}1 & 1\\ 1 & 1\end{bmatrix} = \begin{bmatrix}2 & 2\\ 2 & 2\end{bmatrix}$$

$$A_2^3 = \begin{bmatrix}1 & 1\\ 1 & 1\end{bmatrix}\begin{bmatrix}2 & 2\\ 2 & 2\end{bmatrix} = \begin{bmatrix}4 & 4\\ 4 & 4\end{bmatrix}$$

$$A_2^n = \begin{bmatrix}2^{n-1}\cancel{\cdot(n-1)} & 2^{n-1}\cancel{(n-1)}\\ 2^{n-1}\cancel{(n-1)} & 2^{n-1}\cancel{(n-1)}\end{bmatrix}$$

$$A_3^2 = \begin{bmatrix}a^2 & ab\\ 0 & 0\end{bmatrix} \qquad A_3^3 = \begin{bmatrix}a^3 & a^2b\\ 0 & 0\end{bmatrix} \qquad A_3^n = \begin{bmatrix}a^n & (a^{n-1})b\\ 0 & 0\end{bmatrix} = a^{n-1}\begin{bmatrix}a & b\\ 0 & 0\end{bmatrix}$$

25.

$$\begin{bmatrix}a_{11} & a_{12} & a_{13}\\ a_{21} & a_{22} & a_{23}\\ a_{31} & a_{32} & a_{33}\end{bmatrix}\cdot\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix} = \begin{bmatrix}a_{11}\\ a_{21}\\ a_{31}\end{bmatrix}\begin{bmatrix}1 & 0 & 0\end{bmatrix} + \begin{bmatrix}a_{12}\\ a_{22}\\ a_{32}\end{bmatrix}\begin{bmatrix}0 & 1 & 0\end{bmatrix} + \begin{bmatrix}a_{13}\\ a_{23}\\ a_{33}\end{bmatrix}\begin{bmatrix}0 & 0 & 1\end{bmatrix}$$

$$= \begin{bmatrix}a_{11} & 0 & 0\\ a_{21} & 0 & 0\\ a_{31} & 0 & 0\end{bmatrix} + \begin{bmatrix}0 & a_{12} & 0\\ 0 & a_{22} & 0\\ 0 & a_{23} & 0\end{bmatrix} + \begin{bmatrix}0 & 0 & a_{31}\\ 0 & 0 & a_{32}\\ 0 & 0 & a_{33}\end{bmatrix}$$

26.

$$\begin{bmatrix}1\\ 2\\ 2\end{bmatrix}\begin{bmatrix}3 & 3 & 0\end{bmatrix} + \begin{bmatrix}0\\ 4\\ 1\end{bmatrix}\begin{bmatrix}1 & 2 & 1\end{bmatrix} = \begin{bmatrix}3 & 3 & 0\\ 6 & 6 & 0\\ 6 & 6 & 0\end{bmatrix} + \begin{bmatrix}0 & 0 & 0\\ 4 & 8 & 4\\ 1 & 2 & 1\end{bmatrix} =$$

$$= \begin{bmatrix}3 & 3 & 0\\ 10 & 14 & 4\\ 7 & 8 & 1\end{bmatrix}$$

27.

$$AB = \begin{bmatrix}x & x & x\\ 0 & x & x\\ 0 & 0 & x\end{bmatrix}\begin{bmatrix}x & x & x\\ 0 & x & x\\ 0 & 0 & x\end{bmatrix} = \begin{bmatrix}x^2 & 2x^2 & 3x^2\\ 0 & x^2 & 2x^2\\ 0 & 0 & x^2\end{bmatrix}$$

$$\text{Row 3 of } A \cdot \begin{cases}\text{column 1 of } B\\ \text{column 2 of } B\end{cases} \quad \text{give } 0.$$

$$AB = \begin{bmatrix}x\\ 0\\ 0\end{bmatrix}\begin{bmatrix}x & x & x\end{bmatrix} + \begin{bmatrix}x\\ x\\ 0\end{bmatrix}\cdot\begin{bmatrix}0 & x & x\end{bmatrix} + \begin{bmatrix}x\\ x\\ x\end{bmatrix}\begin{bmatrix}0 & 0 & x\end{bmatrix} = \begin{bmatrix}x^2 & x^2 & x^2\\ 0 & 0 & 0\\ 0 & 0 & 0\end{bmatrix} + \begin{bmatrix}0 & x^2 & x^2\\ 0 & x^2 & x^2\\ 0 & 0 & 0\end{bmatrix} + \begin{bmatrix}0 & 0 & x^2\\ 0 & 0 & x^2\\ 0 & 0 & x^2\end{bmatrix} =$$

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$$= \begin{bmatrix} x^2 & 2x^2 & 3x^2 \\ 0 & x^2 & 2x^2 \\ 0 & 0 & x^2 \end{bmatrix}$$

28.

$$A = \left[\begin{array}{ccc} a_{11} & a_{12} & a_{13} \\ \hline a_{21} & a_{22} & a_{23} \end{array}\right] \qquad B = \left[\begin{array}{c|c|c|c} b_{11} & b_{12} & b_{13} & b_{14} \\ b_{21} & b_{22} & b_{23} & b_{24} \\ b_{31} & b_{32} & b_{33} & b_{34} \end{array}\right]$$

$$A = \begin{bmatrix} A_{11} \\ A_{21} \end{bmatrix} ; \quad B = \begin{bmatrix} B_{11} & B_{12} & B_{13} & B_{14} & \cancel{B_{15}} \end{bmatrix}$$

$$AB = \begin{bmatrix} A_{11}B_{11} & A_{11}B_{12} & A_{11}B_{13} & A_{11}B_{14} \\ A_{21}B_{11} & A_{21}B_{12} & A_{21}B_{13} & A_{21}B_{14} \end{bmatrix} \cancel{\begin{matrix} A_{11}B_{15} \\ A_{21}B_{15} \end{matrix}}$$

1) Matrix A times columns of B.

$$A \cdot \left[\begin{array}{c|c|c|c} B_{11} & B_{12} & B_{13} & B_{14} \end{array}\right]$$

2) $\left[\begin{array}{c} A_{11} \\ \hline A_{21} \end{array}\right] \cdot B$

3) $\left[\begin{array}{c} A_{11} \\ \hline A_{21} \end{array}\right] \cdot \left[\begin{array}{c|c|c|c} B_{11} & B_{12} & B_{13} & B_{14} \end{array}\right]$

4) $\cancel{A \quad A_{1i} = \begin{bmatrix} a_{1i} \\ a_{1i2} \end{bmatrix}}$ $\quad A_{1i} = \begin{bmatrix} a_{1i} \\ a_{2i} \end{bmatrix} ; \; B_{i1} = \begin{bmatrix} b_{i1} \\ b_{i2} \\ b_{i3} \\ b_{i4} \end{bmatrix}^T$

$$A_{11}B_{11} + A_{12}B_{21} + A_{13}B_{31}.$$

29.

$$E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad E_{31} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -4 & 0 & 1 \end{bmatrix}$$

$$E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -4 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ -4 & 0 & 1 \end{bmatrix}$$

$$EA = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ -4 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 & 0 \\ -2 & 0 & 1 \\ 8 & 5 & 3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ -4 \end{bmatrix} [2 \ 1 \ 0] + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} [-2 \ 0 \ 1] + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} [8 \ 5 \ 3] =$$

$$= \begin{bmatrix} 2 & 1 & 0 \\ 2 & 1 & 0 \\ -8 & -4 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ -2 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 8 & 5 & 3 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 3 \end{bmatrix}.$$

30.

$$E = \left[\begin{array}{c|cc} 1 & 0 & 0 \\ \hline 1 & 1 & 0 \\ -4 & 0 & 1 \end{array}\right] \quad \text{so...} \ \cancel{1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, 0 = [0 \ 0], -\vec{c}/a = -4}$$

$$E = \begin{bmatrix} 1 & \vec{0}_{1\times 2} \\ -\vec{c}/a & I_{2\times 2} \end{bmatrix} \quad \text{where } -\vec{c}/a = \begin{bmatrix} -1 \\ 4 \end{bmatrix}$$

$$A = \begin{bmatrix} a & \vec{b}_{1\times 2} \\ \vec{c}_{2\times 1} & D_{2\times 2} \end{bmatrix} \quad \text{so: } a = 2 \quad \vec{b} = [1 \ 0] \quad \vec{c} = \begin{bmatrix} -2 \\ 8 \end{bmatrix} \quad D = \begin{bmatrix} 0 & 1 \\ 5 & 3 \end{bmatrix}$$

$$D - cb/a = \begin{bmatrix} 0 & 1 \\ 5 & 3 \end{bmatrix} - \begin{bmatrix} -2 \\ 8 \end{bmatrix} [1 \ 0] / 2 = \begin{bmatrix} 0 & 1 \\ 5 & 3 \end{bmatrix} - \begin{bmatrix} -2 & 0 \\ 8 & 0 \end{bmatrix} / 2$$

$$= \begin{bmatrix} 0 & 1 \\ 5 & 3 \end{bmatrix} - \begin{bmatrix} -1 & 0 \\ 4 & 0 \end{bmatrix} = \begin{bmatrix} +1 & 1 \\ 1 & 3 \end{bmatrix} \quad \text{same as in P29.}$$

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31.

$$(A+iB)(\vec{x}+i\vec{y}) = A(\vec{x}+i\vec{y}) + iB(\vec{x}+i\vec{y}) =$$

$$= A\vec{x} + iA\vec{y} + iB\vec{x} - B\vec{y}.$$

$$\begin{bmatrix} A & -B \\ iB & iA \end{bmatrix}\begin{bmatrix} \vec{x} \\ \vec{y} \end{bmatrix} = \begin{bmatrix} A\vec{x} - B\vec{y} \\ B\vec{x} + B\vec{y} \end{bmatrix} \begin{matrix} \to \text{real part} \\ \to \text{imaginary part} \end{matrix}$$

32.

$$X = [\vec{x}_1 \ \vec{x}_2 \ \vec{x}_3]$$

$$A \cdot X = I = [A\vec{x}_1 \ A\vec{x}_2 \ A\vec{x}_3]$$

! An essential thing to keep in mind is that matrix-matrix multiplication is just matrix-vector multiplications applied p times $A_{m\times n} \cdot B_{n\times p}$. What A does to $\vec{b}_1$ will do to all $b_i \ \forall i$ (eg. subtract 2 rows).

33.

$$X = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} ; \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} a_{11}+a_{12}+a_{13} & a_{12}+a_{13} & a_{13} \\ a_{21}+a_{22}+a_{23} & a_{22}+a_{23} & a_{23} \\ a_{31}+a_{32}+a_{33} & a_{32}+a_{33} & a_{33} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\Rightarrow A = \begin{bmatrix} 1 & 0 & 0 \\ -1 & +1 & 0 \\ 0 & -1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \\ 8 \end{bmatrix}$$

$$[A\ \vec{b}] = \left[\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ -1 & 1 & 0 & 5 \\ 0 & -1 & 1 & 8 \end{array}\right]$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} [A\ \vec{b}] = \begin{bmatrix} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 8 \\ 0 & -1 & 1 & 8 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \cdot E_{21} [A\ \vec{b}] = \begin{bmatrix} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 8 \\ 0 & 0 & 1 & 16 \end{bmatrix}$$

$$\Rightarrow \vec{x} = \begin{bmatrix} 3 \\ 8 \\ 16 \end{bmatrix}$$

34. As I've formed $A = x \cdot I_{2\times 2} \quad \forall x \in \mathbb{R}$

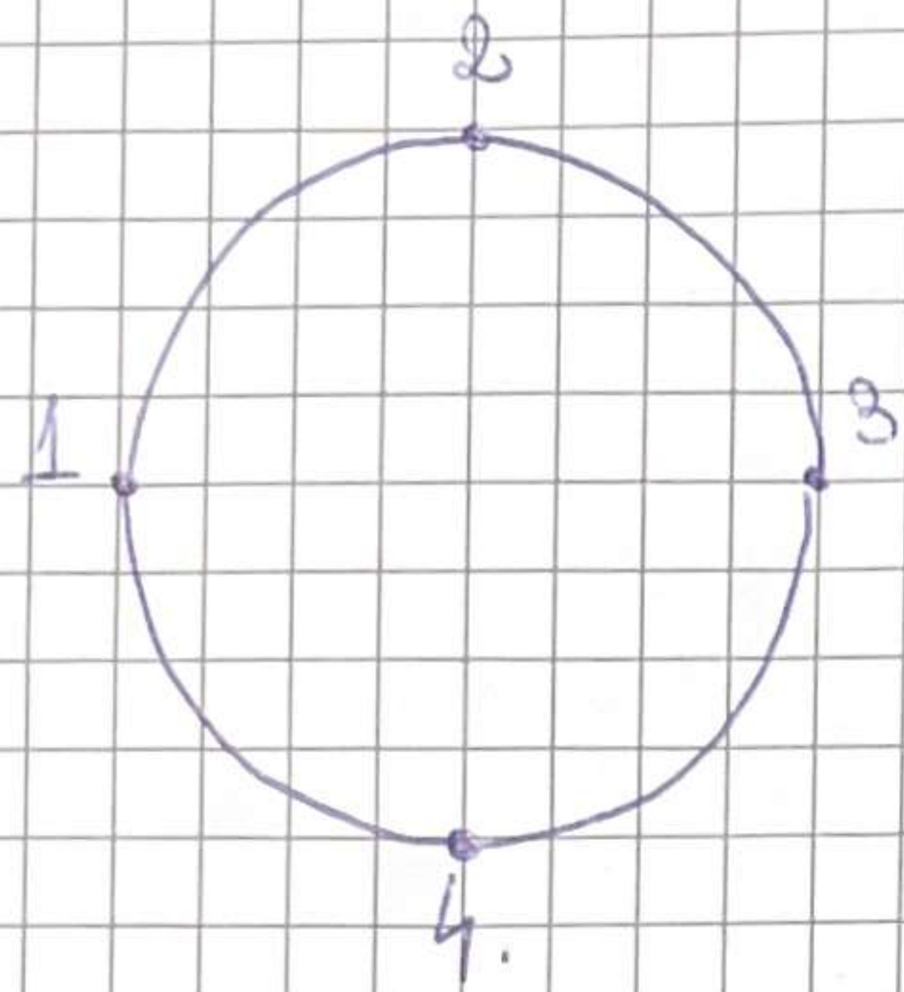
$A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ or $A = x \cdot \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

$$A \cdot \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} a+b & a+b \\ c+d & c+d \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \cdot A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a+c & b+d \\ a+c & b+d \end{bmatrix}$$

$$\begin{cases} a+b = a+c \Rightarrow b = c \\ a+b = b+d \Rightarrow a = d \\ c+d = a+c \Rightarrow a = d \\ c+d = b+d \Rightarrow c = b \end{cases} \Rightarrow \begin{cases} a = d \\ b = c. \end{cases}$$

35.



~~$S =$~~ $S = \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}$

$$S^2 = \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}\begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 2 & 0 \\ 0 & 2 & 0 & 2 \\ 2 & 0 & 2 & 0 \\ 0 & 2 & 0 & 2 \end{bmatrix}$$

~~$1 \to 2 \to 1 \quad 1 \to 4 \to 1$~~

2

$$s_{11} s_{11} + \overset{1}{s_{12} s_{21}} + s_{13} s_{31} + \overset{1}{s_{14} s_{41}}$$

$$s_{11} s_{13} + \underbrace{s_{12} s_{23}}_{1} + s_{13} s_{33} + \underbrace{s_{14} s_{43}}_{1}$$

0
0
0

Challenge Problems

36.

$A_{m \times n}$; $(AB)C \Rightarrow m \cdot n \cdot p + m \cdot p \cdot q$

$B_{n \times p}$ $\quad A(BC) \Rightarrow m \cdot n \cdot q + n \cdot p \cdot q$

$C_{p \times q}$

a) $m = 2$ $\quad mnp + mpq = 2 \cdot 4 \cdot 7 + 2 \cdot \cancel{4}\, 7 \cdot 10 \to$ smaller
$n = 4$
$p = 7$ $\quad mnq + npq = 2 \cdot 4 \cdot 10 + 4 \cdot 7 \cdot 10$
$q = 10$

I prefer $(AB)C$.

b) $(u^T v)\, w^T \to \underbrace{\underbrace{[\quad]_{1\times m}\,[\quad]_{m\times 1}}_{m \text{ multiplications}}\,[\quad]_{1\times n}}_{m + n \text{ multiplications}}$

m

$u^T (v w^T) = \underbrace{[\quad]\,\underbrace{[\quad]\,[\quad]}_{m\times n \text{ multiplications}}}_{m\times n \,+\, m\times n}$

$\Rightarrow (u^T v)\, w^T$ is better.

c) $mnp + mpq / mnpq = np + pq / npq = n + q / nq$

$= \dfrac{n+q}{nq} = \dfrac{n}{nq} + \dfrac{q}{nq} = q^{-1} + n^{-1} = n^{-1} + q^{-1}.$

$mnq + npq / mnpq = \dfrac{m+p}{mp} = m^{-1} + p^{-1}.$

$\Rightarrow (AB)C$ is faster if $n^{-1} + p^{-1} < m^{-1} + p^{-1}$

37. $B = [\vec{b}_1 \ \vec{b}_2 \ \dots \ \vec{b}_m]$

Suppose $C = (c_1, c_2, \dots, c_m)$

columns

If $BA = I$ and $AC = I$, prove $B = C$

$$\left.\begin{array}{c}(BA)C = B(AC) \\ \text{BUT} \\ \begin{cases} BA = I \\ AC = I \end{cases}\end{array}\right\} \Rightarrow C = B$$

38. a) $\cancel{A = [\vec{a}_1 \quad \vec{a}_2 \ldots \vec{a}_m]}$

$$A^T_{m \times m} = [\vec{a}_1^T \quad \vec{a}_2^T \ldots \vec{a}_m^T]$$

$$= \begin{bmatrix} \vec{a}_1 \\ \vec{a}_2 \\ \vdots \\ \vec{a}_m \end{bmatrix}$$

$$A_{m \times m} = [\vec{a}_1 \quad \vec{a}_2 \; --- \; \vec{a}_m^T]$$

$$[\vec{a}_1^T \quad \vec{a}_2^T \ldots \vec{a}_m^T] \cdot \cancel{[\vec{a}_1]} \; [\vec{a}_1 \; \vec{a}_2 \ldots \vec{a}_n]$$

$$= a_1 a_1^T + a_2 a_2^T + \ldots + a_m a_m^T \rightarrow 4^{th} \text{ method}$$

b) $$C = \begin{bmatrix} c_1 & 0 & & & \\ 0 & c_2 & & & \\ & & \ddots & & \\ & & & c_{m-1} & 0 \\ & & & 0 & c_m \end{bmatrix}$$

$\cancel{A^T \cdot C = \vec{a}_1 \cdot [c_1 \; 0 \; 0 \ldots 0]}$

$\cancel{C \cdot A =}$

$\cancel{A^T C A =}$

For $m = 2$:

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} ; A^T = \begin{bmatrix} a_{11} & a_{21} \\ a_{12} & a_{22} \end{bmatrix}$$

OR

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \text{ and } A^T = \begin{bmatrix} a & c \\ b & d \end{bmatrix} \text{ and } C = \begin{bmatrix} x & 0 \\ 0 & y \end{bmatrix}$$

$$A^T A = \vec{a_1} \cdot \vec{a_1}^T + \vec{a_2}\,\vec{a_2}^T = \begin{bmatrix} a \\ b \end{bmatrix} \cdot \begin{bmatrix} a & b \end{bmatrix} + \begin{bmatrix} c \\ d \end{bmatrix} \begin{bmatrix} c & d \end{bmatrix}$$

$$A^T \cdot C = \begin{bmatrix} a & c \\ b & d \end{bmatrix} \begin{bmatrix} x & 0 \\ 0 & y \end{bmatrix} = \begin{bmatrix} ax & cy \\ bx & dy \end{bmatrix} = \cancel{\vec{a_1} \cdot x \cdot c}$$

$$A^T \cdot C \cdot A = \begin{bmatrix} ax & cy \\ bx & dy \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \vec{a_1} \cdot x \cdot \vec{a_1}^T + \vec{a_2} \cdot y \cdot \vec{a_2}^T$$

Then... for m: $A^T C A = \vec{a_1}\, c_1\, \vec{a_1}^T + \vec{a_2}\, c_2\, \vec{a_2}^T + \dots + \vec{a_m}\, c_m\, \vec{a_m}^T$